

User Guide

Beyond Mean-Variance Portfolio Optimiser

With Derivatives & Structured Products — A Mental Accounts Framework

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sami-jeddou-behavioral-portfolio-optimizer.streamlit.app

Overview



Interactive Frontiers

MV vs Behavioural vs Derivative frontier — visualised side by side on a single chart



Three Portfolios

No derivative / With derivative / Same risk — three perspectives for each optimisation run



AI-Powered

On-demand explanations for every parameter, constraint, derivative type, and result

Three Portfolio Perspectives

1

Portfolio (1) — No derivative:

Equivalent to MV optimum — confirms MVT/MAT equivalence. Most classical optimisers stop here.

2

Portfolio (2) — With derivative:

Same implied λ and downside constraint (H, α or H, α, L) — shows return uplift from derivatives.

3

Portfolio (3) — Same risk as (1):

Interpolated at $P(1)$ std dev — direct comparison of return gain at equivalent risk (indicative).

Step 1 — Portfolio Data

1

 **PORTFOLIO DATA**

☐ Default (3-asset sample case)

☒ Live market data (Yahoo Finance)

☐ Enter manually

☐ Upload CSV

Ticker symbols (comma-separated)

AAPL, MSFT, JPM

From

To

2020/01/01

2026/06/03

Return frequency

☒ Daily

☐ Monthly

 Fetch data

Default

Das & Statman (2009) base case — 3 securities, pre-calibrated. Reproduces thesis results.

Live market data

Any global ticker from Yahoo Finance — equities, ETFs, indices, crypto (BTC, ETH...)

Enter manually

Enter means, std devs, and correlation matrix directly. Supports 2–10 securities.

Upload CSV

First col = dates, remaining cols = asset prices. Returns computed automatically.

Step 2 — Derivative / Structured Product

2 DERIVATIVE / STRUCTURED PRODUCT

Safety collar (long put + short call) ▼

None — primary securities only

Put option

Call option

Safety collar (long put + short call)

Aggressive collar (long call + short put)

Straddle (long call + long put)

Strangle (long call + long put, diff strikes)

Capital-guaranteed note — uncapped

2 DERIVATIVE / STRUCTURED PRODUCT

Safety collar (long put + short call) ▼

AI-powered: What is this instrument?

Underlying security
Sec 3 — High risk ▼

Volatility (annualised %)
50.0 - +

Default: 50.0% (std dev of Sec 3 — High risk).
Override for implied vol.

Risk-free rate (%)
3.0 - +

Maturity (years)
1.00
0.25 3.00

Put strike
1.20
0.50 1.50

Call strike
1.60
1.00 2.00

Available instruments

- ▶ Put option / Call option
- ▶ Safety collar (long put + short call)
- ▶ Aggressive collar (long call + short put)
- ▶ Straddle (long call + long put)
- ▶ Strangle (long call + long put, diff strikes)
- ▶ Capital-guaranteed note (capped or uncapped)
- ▶ Barrier-M note
- ▶ Custom composer (calls, puts, digitals, ZCBs)



Select 'None' to optimise without derivatives — Portfolio (1) only

Step 3 — Mental-Account Constraint

3

MENTAL-ACCOUNT CONSTRAINT

Constraint type

☒ VaR — Value at Risk
☐ ES — Expected Shortfall

Threshold H (%)

-40 — -10 — -1

Range extended to -40% to accommodate highly volatile assets (e.g. cryptocurrencies, emerging market equities).

Shortfall probability α (%)

1 — 5 — 15

VaR constraint: $P(\text{return} < H) \leq \alpha$

Implied risk-aversion $\lambda = 3.6958$

MV optimal at $\lambda=3.70 \equiv$ behavioural optimal at $H=-10\%, \alpha=5\%$

AI-powered: What is the VaR constraint?

VaR — Value at Risk

$$P(\text{return} < H) \leq \alpha$$

The probability of the portfolio return falling below threshold H must not exceed α . Set H (loss threshold) and α (max probability) using the sliders.

ES — Expected Shortfall

$$E[\text{return} \mid \text{return} < H] \geq L$$

The expected loss in the worst $\alpha\%$ of scenarios below H must not exceed L. Set H, α , and L using the sliders.

💡 Implied λ is computed dynamically — adjust H and α to see the equivalent risk-aversion coefficient update in real time, confirming MVT/MAT equivalence.

Step 4 — Grid Resolution



The screenshot shows the 'GRID RESOLUTION' step of the optimiser. At the top, a dark blue header contains a blue circle with the number '4' and the text 'GRID RESOLUTION' with a lightning bolt icon. Below this is a dropdown menu showing 'Fast (m=21, m'=15)' with a downward arrow. Under the dropdown is a light blue button with a star icon and the text 'AI-powered: What does this resolution mean?'. At the bottom are two large blue buttons: '5 ► RUN OPTIMISER' and '↶ Reset / New Simulation'.

Fast (m=21, m'=15)

 ~1–2 min

Quick preview and parameter exploration

Standard (m=35, m'=50)

 ~5–15 min

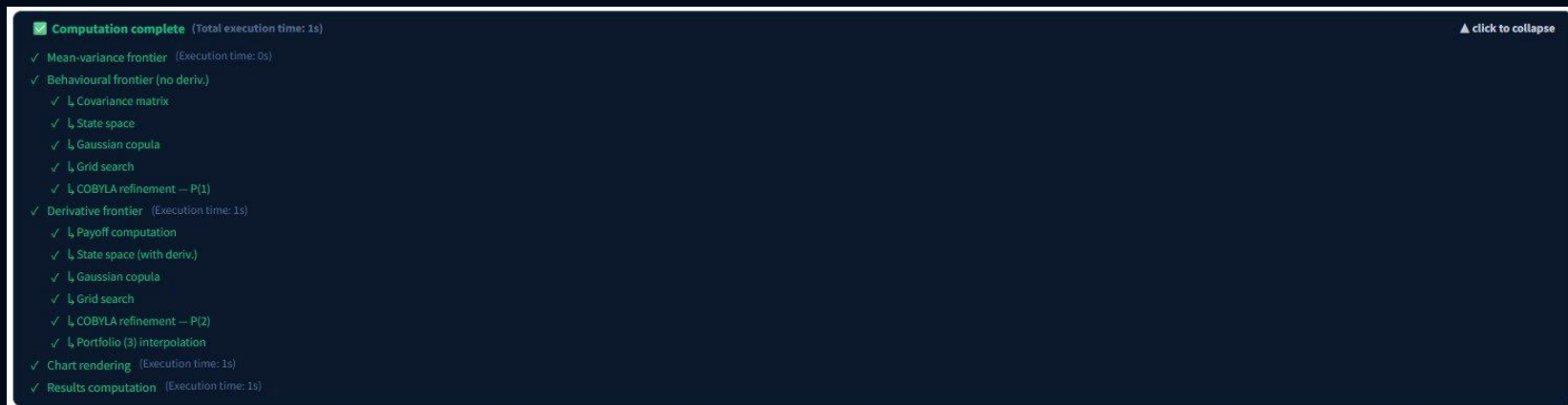
Good balance — close to thesis results

High precision (m=51, m'=99)

 1–2 hrs

Publication-quality replication of thesis results

Step 5 — Run the Optimiser



Computation steps

1. Mean-variance frontier — Markowitz sweep (instant)
2. Behavioural frontier (no derivative) — State space, Gaussian copula, grid search, COBYLA
3. Derivative frontier — Same steps extended with derivative payoff (only if derivative selected)
4. Chart rendering + Results computation

Reading the Chart

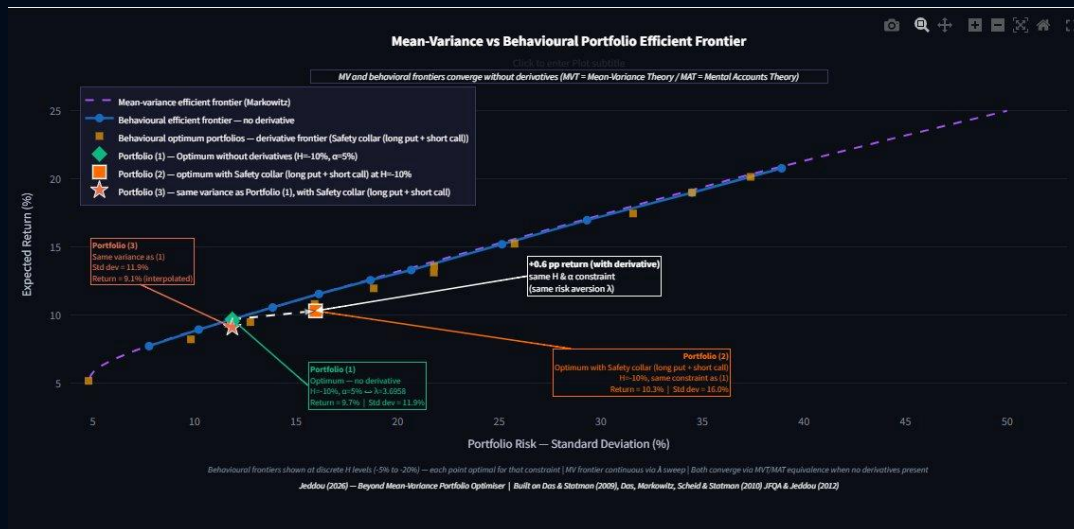
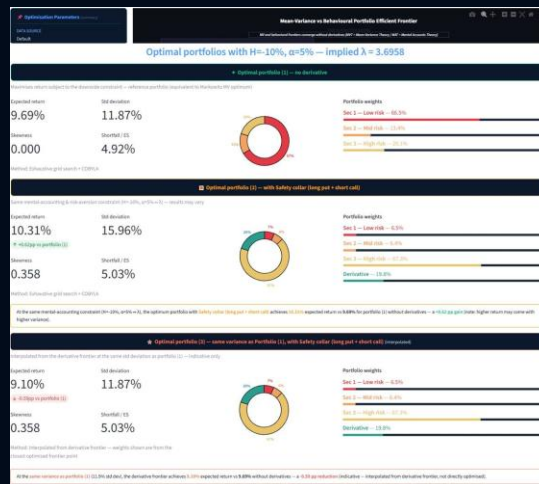


Chart legend

- Purple dashed**
Mean-variance efficient frontier (Markowitz)
- Blue dots**
Behavioural efficient frontier — no derivative
- Gold squares**
Behavioural optimum — derivative frontier
- Green diamond**
Portfolio (1) — no derivative optimum
- Orange square**
Portfolio (2) — with derivative, same λ
- Coral star**
Portfolio (3) — same variance as P(1)

X-axis: Portfolio Risk (Standard Deviation %) | Y-axis: Expected Return (%)

Portfolio Results



Key metrics

Expected return: Annualised mean return of the optimal portfolio

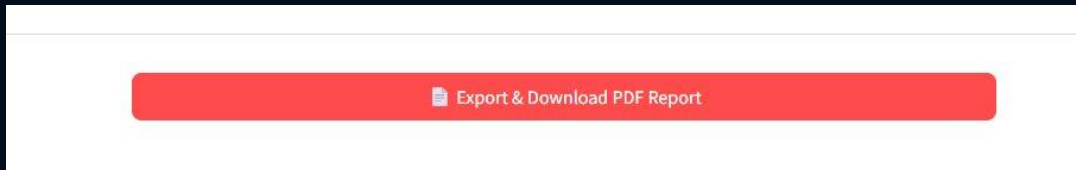
Shortfall / ES: Expected loss in the tail (ES constraint value)

Std deviation: Annualised portfolio volatility (risk measure)

Portfolio weights: Allocation to each security and derivative

Skewness: Asymmetry of return distribution — positive = upside skew

PDF Export



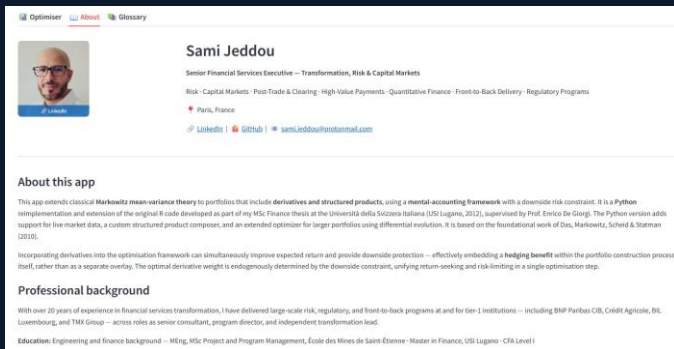
PDF report includes

- ✓ Cover page with run parameters
- ✓ Full efficient frontier chart
- ✓ Portfolio (1), (2), (3) metrics
- ✓ Weight allocation bar charts
- ✓ Legal disclaimer

💡 The PDF is generated in-browser — no data is sent externally. Click 'Download PDF report' after the simulation completes.

About & Glossary Tabs

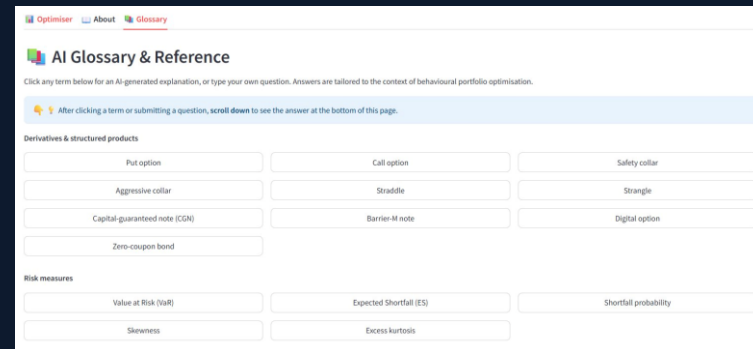
About Tab



The screenshot shows the 'About' tab of the Optimiser application. It features a profile for Sami Jeddou, a Senior Financial Services Executive. The page includes a bio, a list of skills (Risk, Capital Markets, Post-Trade & Clearing, High-Value Payments, Quantitative Finance, Front-to-Back Delivery, Regulatory Programs), and social media links for LinkedIn, GitHub, and email. Below this is a section titled 'About this app' which describes the app as a Python implementation of the original R code, supporting live market data and a custom structured product composer. It also mentions that incorporating derivatives into the optimisation framework can improve expected return and provide downside protection. A 'Professional background' section follows, detailing Sami's 20+ years of experience in financial services transformation, including roles at BNP Paribas CIB, Credit Agricole, BIL, Luxembourg, and TMO Group. The page concludes with an 'Education' section listing his MEng, MSc, and PhD degrees.

- ▶ Author background & contact
- ▶ App description & algorithm
- ▶ Academic references
- ▶ Legal disclaimer

Glossary Tab



The screenshot shows the 'Glossary' tab of the Optimiser application. It is titled 'AI Glossary & Reference' and includes a sub-header 'Click any term below for an AI generated explanation, or type your own question. Answers are tailored to the context of behavioural portfolio optimisation.' Below this is a light blue box with a tip: 'After clicking a term or submitting a question, scroll down to see the answer at the bottom of this page.' The main content area is divided into two sections: 'Derivatives & structured products' and 'Risk measures'. Each section contains a grid of buttons for various financial terms. In the 'Derivatives & structured products' section, the buttons are: Put option, Call option, Safety collar, Aggressive collar, Straddle, Strangle, Capital-guaranteed note (CGN), Barrier-M note, Digital option, and Zero-coupon bond. In the 'Risk measures' section, the buttons are: Value at Risk (VaR), Expected Shortfall (ES), Shortfall probability, Skewness, and Excess kurtosis.

- ▶ Click any term for AI explanation
- ▶ Covers derivatives, risk, portfolio theory
- ▶ Ask your own questions
- ▶ Tailored to behavioural portfolio context

Disclaimer & Contact

Legal Disclaimer

This application is based on the mental accounts portfolio optimisation framework of Das & Statman (2009) and Das, Markowitz, Scheid & Statman (2010), as extended in Jeddou (2012). It is provided for educational and research purposes only and does not constitute financial advice, investment recommendations, or a solicitation to buy or sell any financial instrument. Results are purely illustrative and should not be used as the basis for any investment decision.

Academic References

- ▶ Das, S. & Statman, M. (2009) — Beyond Mean-Variance: Portfolios with Derivatives and Non-Normal Returns in Mental Accounts
- ▶ Das, S., Markowitz, H., Scheid, J. & Statman, M. (2010) — Portfolio Optimization with Mental Accounts, JFQA Vol.45 No.2
- ▶ Jeddou, S. (2012) — Beyond Mean-Variance: Options and Structured Products in Behavioral Portfolios, MSc Finance, USI Lugano



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